



Synthesis and Nuclear Radiation Shielding Ability of $\text{Li}_2\text{O-ZnO-P}_2\text{O}_5$ Glasses: The Role of Yb_2O_3

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Abstract

Nuclear radiation shielding parameters for glass samples composed of 20 mol.% Li_2O -40 mol.% ZnO -40 mol.% P_2O_5 : x wt.% Yb_2O_3 ($x=0, 0.2, 0.4, 0.6, 0.8, 1, 2$) were evaluated. The samples were synthesized using the melt-quenching technique. The prepared samples were encoded Yb0.0, Yb0.2, Yb0.4, Yb0.6, Yb0.8, Yb1 and Yb2 according to the Yb_2O_3 content. X-ray diffraction (XRD) reveals that all the prepared samples were amorphous. The gamma ray shielding parameters such as mass attenuation coefficient (MAC), linear attenuation coefficient (LAC), half value layer (HVL) and mean free path (MFP) were obtained using Phy-X/PSD and XCOM software programs. The atomic cross section (ACS), electronic cross section (ECS), effective atomic number (Z_{eff}), and electron density (N_{eff}) were evaluated, as well as exposure buildup factor (EBF) and energy absorption buildup factor (EABF). The MAC values were found to be 26.304–31.872 cm^2/g for Yb0.0–Yb2, respectively. The HVL and MFP values were found to be (0.008–0.007 cm) and (0.012–0.010 cm) for Yb0.0–Yb2, respectively. The ACS and ECS values were found to be $(9.912 \times 10^{-21} \text{ cm}^2/\text{g})$ up to $13.06 \times 10^{-21} \text{ cm}^2/\text{g}$ and (3.990×10^{-23}) up to $4.900 \times 10^{-23} \text{ cm}^2/\text{g}$ for Yb0.0–Yb2, respectively. The Z_{eff} and N_{eff} values were found to be (24.83–26.64) and (6.59×10^{23}) up to 6.50×10^{23} electrons/g for Yb0.0–Yb2, respectively. The results showed that the addition of Yb_2O_3 improves gamma-ray shielding properties of the glass. On the other hand, adding Yb_2O_3 to the prepared glass leads to a decrease in the removal cross section for fast neutrons (Σ_R) from 0.103 to 0.096 cm for Yb0.0–Yb2, respectively. The gamma-ray shielding parameters for the present samples were compared with commercial concrete and glass used as gamma ray shields.

Keywords Phosphate glass · gamma ray shields · Yb · nuclear parameters

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Introduction

Recently, many studies have been published related to the use of doped glass for radiation protection. It has been proven that glass can be an effective material to protect against gamma rays due to the possibility of doping it with heavy elements such as lead, bismuth, barium and other elements that are used as gamma-ray shields. Some nuclear shielding parameters such as mass attenuation coefficient, half-value layer, mean free path, effective electron density, effective atomic number as well as atomic and electron cross sections and dose coefficients were used to evaluate the shielding properties of the prepared glass.^{1–4}

Phosphate (P_2O_5) glass is considered the most exciting and attractive optical glass due to its unique properties, such as low melting point, high durability, and high UV transmittance. The addition of rare earth elements (REE) to P_2O_5 glasses gives the composite glasses high transparency, and thus is used in optical applications. Phosphate glass doped with trace element oxides such as ytterbium oxide have made great progress in obtaining fiber lasers with improved optical and radiation shielding properties.⁵ Moreover, adding oxides such as zinc oxide (ZnO) to the main component of the glass plays an important role as it acts as a modifier and improves the shielding properties.⁶

Several authors have made more effort to prepare and study the shielding properties of many types of glasses, as well as measuring their mechanical, optical and thermal properties.^{6–8} Rammah et al.⁹ studied the effect of Gd_2O_3 on the physical properties of borate bismuth glass, and measuring their shielding properties for neutron and gamma radiation. They found that the addition of Gd_2O_3 to the borate bismuth glass improves the shielding properties of the prepared glass. Alshahrani et al.¹⁰ studied the shielding properties of tellurite glass containing Sb_2O_3 and V_2O_5 . The results showed that the addition of antimony oxide improves the gamma shielding properties, as well as the fast neutron removal cross section. Mariselvam¹¹ studied radiation shielding features of barium bismuth fluoroborate glass doped with Yb^{3+} . The results indicated that increasing the content of ytterbium dioxide improved the shielding properties of the prepared glass. Rammah et al.² studied gamma-ray features for bismuth borate glasses doped with various concentrations of Yb_2O_3 . The results indicated that the addition of Yb_2O_3 to the prepared glasses led to improve gamma ray shielding properties. Naseer et al.¹² studied the impact of Yb_2O_3 on bismuth boro-phosphate glass. The results indicated that the addition of Yb_2O_3 leads to an improvement in the optical properties as well as the gamma-ray shielding properties. Saleh et al.¹³ studied the nuclear radiation shielding of LNBP glass doped with the increase of $CdO + Te$. The

results showed an improvement in the nuclear radiation shielding with an increase of $CdO + Te$. Algradee et al.¹⁴ synthesized lithium-zinc-phosphate glasses doped with Nd^{+3} and evaluation of nuclear radiation shielding for the prepared glass system. The results showed that the nuclear shielding parameters are improved by adding Nd_2O_3 to the prepared glass system. Saleh et al.¹⁵ fabricated a novel lithium lead bismuth borate glass system for nuclear radiation shielding. The results indicated that the addition of Bi_2O_3 increased the mass attenuation coefficient (MAC) and effective atomic number (Z_{eff}) and reduced the half-value layer (HVL), mean free path (MFP), and buildup factors. Alwany et al.¹⁶ studied structural, optical and radiation shielding properties of ZnS nanoparticles QDs. The results indicated that the LAC is increased with increasing nanoparticle size, while HVL, total value layer (TVL) and MFP are decreased as the particle size decreases.

The Phy-X/PSD software program is an easy-to-use, online radiation shielding parameter measurement and dosimeter, available at <https://phy-x.net/PSD>.¹⁷ These parameters include all nuclear radiation parameters such as LAC, MAC, MFP, HVL, TVL, Z_{eff} , N_{eff} together with energy exposure buildup factor (EABF) and exposure buildup factor (EBF). The program gives the values of the aforementioned factors in the energy range from 1 keV up to 100 GeV. In addition, with this program, it is possible to calculate the fast neutron removal cross section (FNRCs) of a compound or mixture. Also, the XCOM program is used to find the mass attenuation coefficient for all elements, compounds, or mixtures within the energy range of 1–100 GeV.¹⁸

The current work aims to evaluate the nuclear radiation shielding for prepared $ZnO-Li_2O-P_2O_5$ glass doped with various concentrations of Yb_2O_3 . The effect of Yb_2O_3 on the radiation shielding properties of gamma rays and fast neutron removal cross section were studied using XCOM and Phy-X/PSD software programs for photon energies from 0.015 up to 15 MeV. Moreover, the structural properties of the prepared samples using x-ray diffraction were evaluated.

Materials and Methods

Samples Preparation

The conventional technique (melt-quenching) was applied to fabricate glasses with composition of 20 mol.% Li_2O –40 mol.% ZnO –40 mol.% P_2O_5 : x wt.% Yb_2O_3 ($x=0, 0.2, 0.4, 0.6, 0.8, 1, 2$). The prepared glasses are encoded as Yb0.0, Yb0.2, Yb0.4, Yb0.6, Yb0.8, Yb1 and Yb2 depending on the concentration of Yb_2O_3 in the prepared glass. The high purity of all the oxides used in the composition of the presented glass system was weighed, grinding and well mixed, then pre-heated at a temperature of 300°C for 1 h. The weighted mixtures were

placed in a silica crucible and placed in a furnace for melting at 1100°C for 1 h. Then the mixture was poured into molds of suitable size. The produced glass was transferred to a melting furnace at 300°C for the annealing process. Samples were periodically swirled every 10 min to ensure their homogeneity. The glass samples were kept in a drying furnace for 1 day to prevent moisture entry into the samples. Figure 1 shows the schematic description of the glass preparation.

To confirm the nature of the prepared samples, the samples were examined by x-ray diffraction measurements using a Bruker D8 Discover x-ray diffractometer, Germany. All measurements were made with a wavelength of 1.54 Å of Cu Kα.

The Archimedes principle was employed to measure the glass density, ρ_g , using toluene as a liquid medium ($\rho_{\text{toluene}} = 0.864 \text{ g cm}^{-3}$), according to the following relation:

$$\rho_g = \left[\rho_{\text{toluene}} \frac{(m_{g,a.} - m_{l.a.})}{(m_{g.a.} - m_{l.a.}) - (m_{g,t.} - m_{l.t.})} \right] \quad (1)$$

where $m_{g,t.}$ and $m_{g,a.}$ are the masses of the glass in toluene and air, while $m_{l,t.}$ and $m_{l,a.}$ are masses of suspended thread

in toluene and air, respectively. The accuracy was $\pm 0.001 \text{ g cm}^{-3}$. Molar volume, V_m was computed by Eq. (2):

$$V_m = \frac{\text{Molecular weight of the glass sample}}{\rho_g} \quad (2)$$

High-purity raw materials suitable for optical glasses were used. Laboratory reagent-grade annular ZnO (Loba Chemie, 99.5%), Li₂CO₃ (Loba Chemie, 99%), P₂O₅ (Ottokemi, 99.99%) and Yb₂O₃ (Sigma-Aldrich, 99.9%) were weighed in appropriate amounts.

The samples were measured firstly at room temperature (25°C) in the air and then in the buoyant liquid (toluene) using an electronic balance with a sensitivity of (0.0001). Table I shows the density of the present samples as well as the percentages of each element.

Computational Work

Mass attenuation coefficients (MAC) of prepared glass samples encoded Yb0.0, Yb0.2, Yb0.4, Yb0.6, Yb0.8,

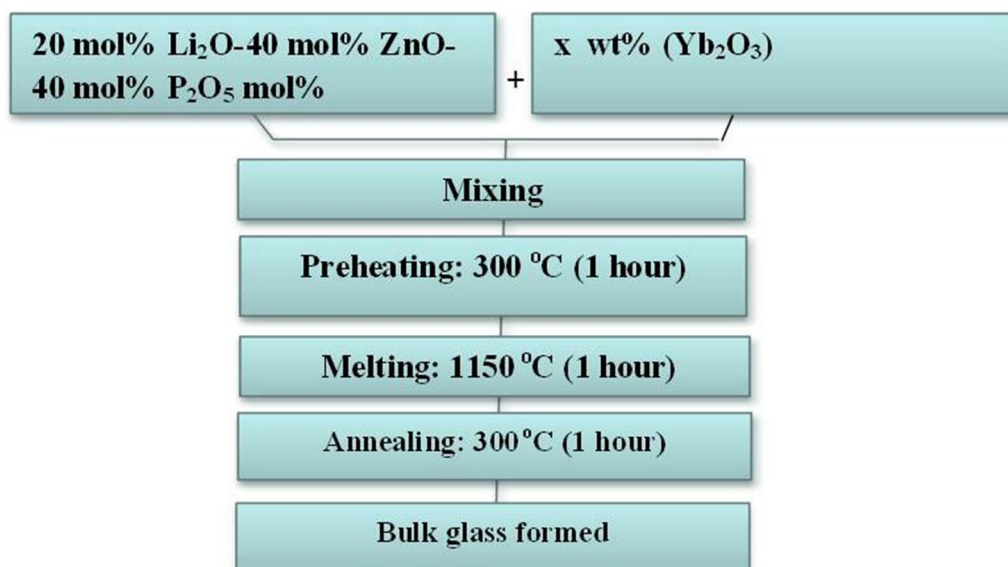


Fig. 1 Schematic description of the glass preparation.

Table I Codes, chemical compositions (mol), molecular weight (MW) and density of the prepared glass samples.

Code	ZnO	P ₂ O ₅	Li ₂ O	Yb ₂ O ₃	MW, (g/mol)	Density (g/cm ³)
Yb0.0	0.4	0.4	0.2	0	95.304	3.1551
Yb0.2	0.399797	0.3997971	0.1998985	0.000507	95.45556	3.1083
Yb0.4	0.399594	0.3995944	0.1997972	0.001014	95.60696	3.1054
Yb0.6	0.399392	0.3993919	0.199696	0.00152	95.7582	3.1034
Yb0.8	0.39919	0.3991896	0.1995948	0.002026	95.9093	3.1004
Yb1	0.398988	0.3989875	0.1994938	0.002531	96.06024	3.0913
Yb2	0.39798	0.3979802	0.1989901	0.005049	96.81266	3.1225

Yb1 and Yb2 according to 20 mol.% Li₂O-40 mol.% ZnO-40 mol.% P₂O₅:*x* wt% Yb₂O₃ (*x*=0, 0.2, 0.4, 0.6, 0.8, 1, 2) were evaluated using XCOM¹⁴ and Phy-X/PSD¹⁵ software programs.

Based on the values of MAC, the linear attenuation coefficient (LAC), the half value layer (HVL), the mean free path (MFP), the effective atomic number (Z_{eff}), electron density (N_e), atomic cross section (ACS), electronic cross section (ECS), energy absorption buildup factor (EABF), exposure buildup factor (EBF) and effective removal of neutron cross section (Σ_R) were calculated. Theoretical concepts and equations of the coefficients are as follows:

The mass attenuation coefficient (cm²/g) for any material is obtained by dividing the linear attenuation coefficient by the density of the substance according to the following equation¹³:

$$\text{MAC} = \frac{\mu}{\rho} \quad (3)$$

where ρ is the material density (g/cm³).

The half value layer (HVL) of any given material is defined as the thickness of the material that attenuates 50% of incident radiations. HVL is expressed in units of distance (cm) and is given by the following relation¹⁴:

$$\text{HVL(cm)} = \frac{0.693}{\mu} \quad (4)$$

The mean free path (MFP) is the distance traveled by a moving particle or photon between two successive collisions with other particles, and it can be given by the following relation:

$$\text{MFP(cm)} = \frac{1}{\mu} \quad (5)$$

The atomic cross section (ACS) can be easily determined from the following equation¹⁹:

$$\text{ACS} = \frac{1}{N_A} \sum_i f_i N_i \left(\frac{\mu}{\rho} \right)_i \quad (6)$$

where f_i is the fractional abundance of element *i* concerning the number of atoms;

A_i is the atomic weight, respectively, of the constituent element *i*. The electronic cross section (ECS) for the individual element is expressed by the following formula¹⁹:

$$\text{ECS} = \frac{1}{N_i} \sum_i \frac{f_i A_i}{Z_i} \left(\frac{\mu}{\rho} \right)_i \quad (7)$$

The effective atomic number (Z_{eff}) of a substance has a strong and fundamental relationship with the nature of radiation interactions within this medium. For any compound substance consisting of more than one element, the Z_{eff} must be specified for that substance. The effective atomic number (Z_{eff}) of the compound is given by the following equation¹⁵:

$$Z_{\text{eff}} = \frac{\sum_i f_i A_i \left(\frac{\mu}{\rho} \right)_i}{\sum_j \frac{A_j}{Z_j} \left(\frac{\mu}{\rho} \right)_j} \quad (8)$$

where f_i is a molar fraction, A_i is atomic mass and Z_i is the atomic number.

The effective electron number or electron density (N_e) (number of electrons per unit mass), can be derived from¹⁴:

$$N_e = \frac{\mu/\rho}{\text{ECS}} \quad (9)$$

Energy absorption buildup factor (EABF) and exposure buildup factor (EBF) values are useful in selecting an alternative composite material rather than an element at the required energy. The EBF and EABF values for a protective layer of gamma rays, with a time of up to 40 mfp and energy of 0.015 up to 15 MeV, are calculated using the geometric progression (G-P) fitting parameters (*b*, *c*, *a*, x_k and *d*) according to the following equations¹⁶:

$$B(E, X) = 1 + \frac{(b-1)(k^x - 1)}{k-1} \text{ for } k \neq 1 \quad (10)$$

$$k(E, X) = cx^a + d \frac{\tanh\left(\frac{x}{x_k} - 2\right) - \tanh(-2)}{1 - \tanh(-2)} \quad (11)$$

$$B(E, X) = 1 + (b-1)x \text{ for } k = 1 \quad (12)$$

where *x* is the distance in the medium in terms of MFP, *K* is the dose multiplicative factor, and *b*, *c*, *a*, x_k , and *d* are G-P fitting parameters.

Effective removal of neutron cross section Σ_R :

The effective removal of neutron cross section refers to the process of removing neutrons from materials at the appropriate time. The Σ_R values are a measure of the interaction probability between the fast neutrons and the target. The Σ_R values for a substance can be calculated using the following equation¹⁶:

$$\Sigma_R = \Sigma w_i \left(\Sigma_{R/\rho} \right) \quad (13)$$

where w_i is partial density (g/cm³) of the *i*th element and $\Sigma_{R/\rho}$ is mass removal cross section (cm²/g).

Results and Discussion

The density of the present glasses Yb0.0–Yb2 is shown in Table I. It is clear from the table that the density of the present glasses decreases with the increasing percentage of Yb_2O_3 except for sample Yb2. The anomalous behavior of density may be because of two opposing factors. The first one is related to the introduction of Yb_2O_3 with higher molecular weight into the glass matrix (The molecular weight of Yb_2O_3 (394.08 g/mol) is heavier than the molecular weight of other oxides in the glasses under study and the molecular weight of ZnO , P_2O_5 and Li_2O are 81.38, 141.94 and 29.88 g/mol, respectively), while the other factors are related to the formation of non-bridging oxygen in the network due to the ionic radius of Yb^{3+} (0.87 Å) are greater than the ionic radii of other cations in the present glasses composition (the ionic radii of Zn^{2+} , P^{5+} and Li^+ are 0.74, 0.17 and 0.59 Å, respectively).

The decrease in the density of the samples with the increase in the content of Yb_2O_3 in the compositional region (0–1 wt.%) is related to the formation of non-bridging oxygen within the glass matrix. The increase of density for sample Yb2 due to the increase in the content of Yb_2O_3 leads to the filling of the interstitial spaces in the glass network.

Figure 2 shows the XRD measurements of prepared samples with different contents of Yb_2O_3 . These patterns do not show any clear peaks, and thus proved that the prepared glass samples were in the form of an amorphous phase. The MACs of the Yb0.0–Yb2 glasses were computed using the Phy-X/PSD software program and compared with that obtained from the online XCOM software program in the photon energy range from 15 keV up to 15 MeV and the results are given in Table II.

Figure 3 shows the variance in the values of MACs for the Yb0.0–Yb2 glass samples with energy. It can be seen from the figure that the highest MAC values for the present samples were seen at low photon energy. Then, as the photon's energy increases, the MAC values undergo a sharp decrease. The increase in MAC values in this lower energy region is due to the photoelectric absorption (PEA) prevalent in this region and has a cross section proportional to $Z^{4-5}/E^{3.5}$. In the energy region near 0.1 MeV, the Compton effect is dominant and its cross section is proportional to Z/E . In the high-energy region, the MAC values are close to zero and exhibit quasi-constant behavior with photon energy for all glass samples, and the pair production (PP) is the dominant process where the cross section is proportional to Z^2 . The MAC values for the presented samples Yb0.0–Yb2 at the energy of 15 keV were found to be 26.304–31.872 cm^2/g , respectively. This means that the addition of Yb_2O_3 leads to an increase in the MAC

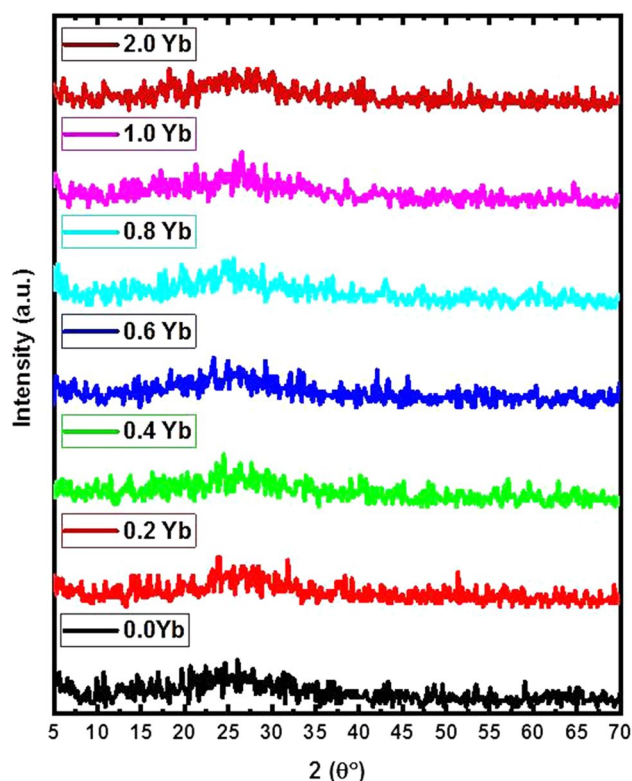


Fig. 2 XRD patterns for the prepared glass samples.

due to the higher atomic number compared with the other elements in the prepared glass samples.

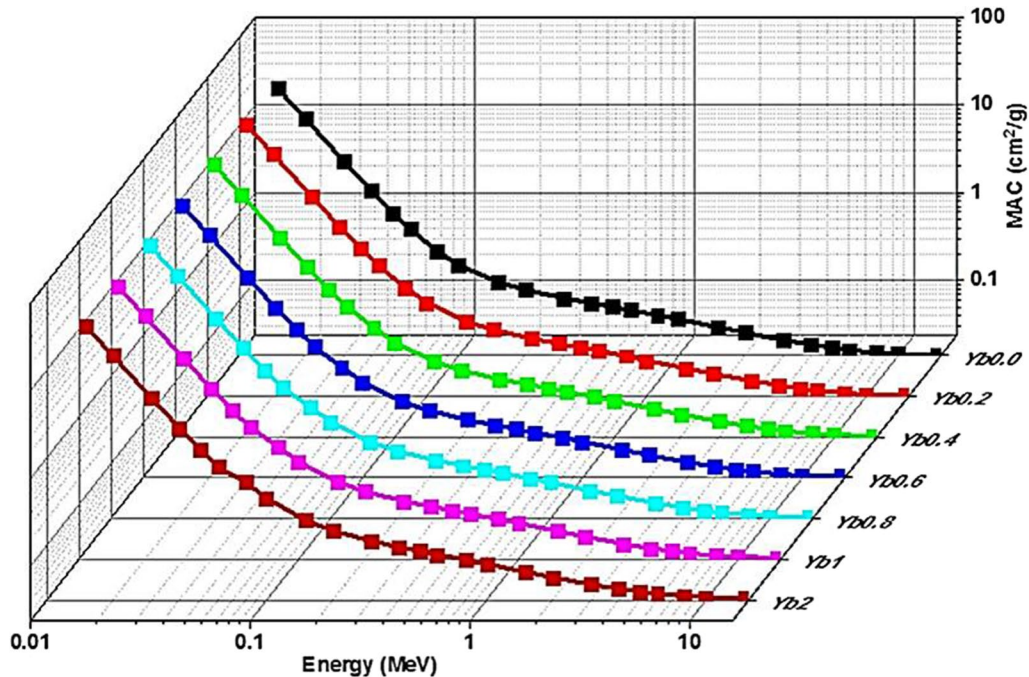
Figure 4 shows the obtained results of the MAC for Yb2 present glass sample compared with the corresponding values for ordinary and barite concrete, as well as some commercial glass RS-360 and RS-253 G18. It should be noted that the values of the MAC for the Yb2 present sample are higher than the MACs for ordinary concrete (OC) and RS-252 G18 glass and less than MACs for barite concrete (BaC) and RS-360 glass.

Figure 5 shows the values of LAC versus photon energy values in the range from 15 keV to 15 MeV. From the figure, it should be noted that the LAC values ranged between 82.992 cm^{-1} and 99.520 cm^{-1} for samples Yb0.0 and Yb2, respectively. Also, it is clear from the results that the behavior of the LAC is like the behavior of the MAC with the increase in the energy of the photon.

Figure 6 shows the HVL values for the present samples versus photon energy. The HVL is an important feature of the material that describes the thickness of the material, which reduces the intensity of the incident gamma rays by half value. From the figure, the HVL values increase with increasing photon energy for all present samples. Also, it can be noted that the highest value was for sample Yb0.0 (0.008 cm) and the lowest value was for sample Yb2 (0.007 cm), and this reveals that adding the Yb_2O_3 led to

Table II Mass attenuation coefficients from Phy-X/PSD and XCOM programs.

Energy (MeV)	Yb0.0		Yb0.2		Yb0.4		Yb0.6		Yb0.8		Yb1		Yb2	
	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM
0.015	26.304	26.310	30.281	29.550	30.461	29.790	30.641	29.930	30.819	30.140	30.997	30.280	31.872	31.190
0.02	11.982	11.980	13.793	13.480	13.879	13.590	13.965	13.660	14.050	13.760	14.135	13.830	14.554	14.270
0.03	3.923	3.924	4.503	4.411	4.534	4.450	4.564	4.474	4.593	4.509	4.623	4.533	4.770	4.686
0.04	1.805	1.806	2.060	2.021	2.074	2.039	2.088	2.051	2.102	2.067	2.116	2.079	2.186	2.151
0.05	1.019	1.019	1.152	1.132	1.160	1.143	1.168	1.149	1.176	1.158	1.184	1.165	1.222	1.205
0.06	0.661	0.661	0.739	0.728	0.744	0.734	0.749	0.738	0.754	0.744	0.759	0.748	0.783	0.773
0.08	0.367	0.367	0.412	0.405	0.426	0.422	0.439	0.434	0.453	0.451	0.466	0.462	0.533	0.530
0.1	0.256	0.257	0.280	0.276	0.287	0.286	0.295	0.292	0.302	0.302	0.310	0.308	0.347	0.346
0.15	0.164	0.164	0.171	0.170	0.174	0.174	0.176	0.176	0.179	0.179	0.181	0.181	0.194	0.194
0.2	0.134	0.134	0.137	0.136	0.138	0.138	0.139	0.139	0.140	0.140	0.141	0.141	0.147	0.147
0.3	0.108	0.108	0.109	0.109	0.109	0.109	0.109	0.110	0.110	0.110	0.110	0.110	0.112	0.112
0.4	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.096	0.096
0.5	0.086	0.086	0.085	0.086	0.086	0.086	0.086	0.086	0.086	0.086	0.086	0.086	0.086	0.086
0.6	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079
0.8	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069	0.069
1	0.062	0.062	0.061	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062
1.5	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
2	0.044	0.044	0.043	0.043	0.043	0.043	0.043	0.043	0.043	0.043	0.043	0.043	0.043	0.043
3	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036
4	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032
5	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.030	0.029	0.030	0.029	0.030	0.030
6	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028
8	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026
10	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.026	0.025
15	0.024	0.024	0.025	0.024	0.025	0.024	0.025	0.024	0.025	0.025	0.025	0.025	0.025	0.025

**Fig. 3** MAC versus photon energy for all glass samples.

a decrease in the HVL values. Figure 7 depicts the comparison of the values of the HVL of the present samples with the corresponding values for the ordinary concrete and barite, as well as commercial glass known as RS-360 and RS-253 G18. It should be noted that the values of the

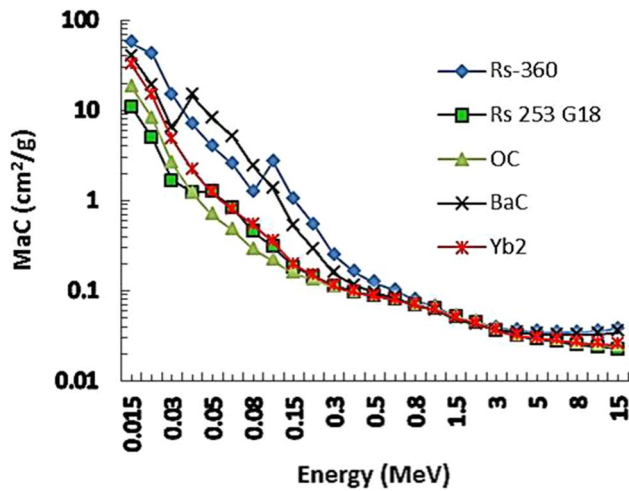


Fig. 4 Comparison of the minimum and maximum values of the MAC of the Yb2 and commercial materials.

present samples are lower than the ordinary concrete (OC) and RS-253 G18 glass and higher than the barite concrete (BaC) and RS-360 glass.

The mean free path (MFP) is the distance traveled between two successive collisions of gamma ray photons. Materials with a lower MFP are better than those with a higher MFP in terms of their use as gamma ray shields. Figure 8 shows the values of the MFP for the present samples versus the photon energy. From the figure, it can be seen that the values of the MFP increase with the increase in the energy of the photon. It was also noticed that the lowest values were for sample Yb2 while the higher values were for sample Yb0.0. At an energy of 15 keV, the MFP value for sample Yb0.0 was 0.012 cm, while for sample Yb2 it was 0.010 cm. Also, it is noticed that the general behavior of MFP is similar to that of HVL.

The cross section in physics defines that a given atomic nucleus or a subatomic particle will exhibit a specific interaction (for example, absorption or scattering) with the incident photon. Figures 9 and 10 show the behavior of atomic and electronic cross sections (ACS and ECS) with photon energy. It is clear from Figs. 8 and 9 that the two parameters tend to have the same behavior with the photon energy. At an energy of 15 keV, the ACS value for sample Yb0.0 was

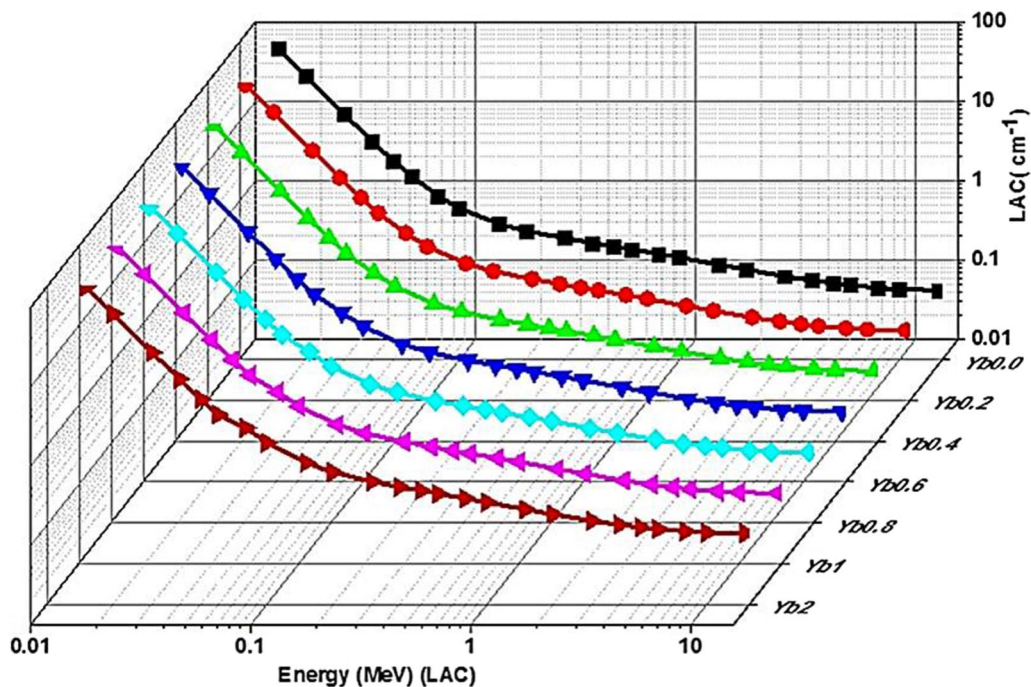


Fig. 5 LAC versus photon energy for all glass samples.

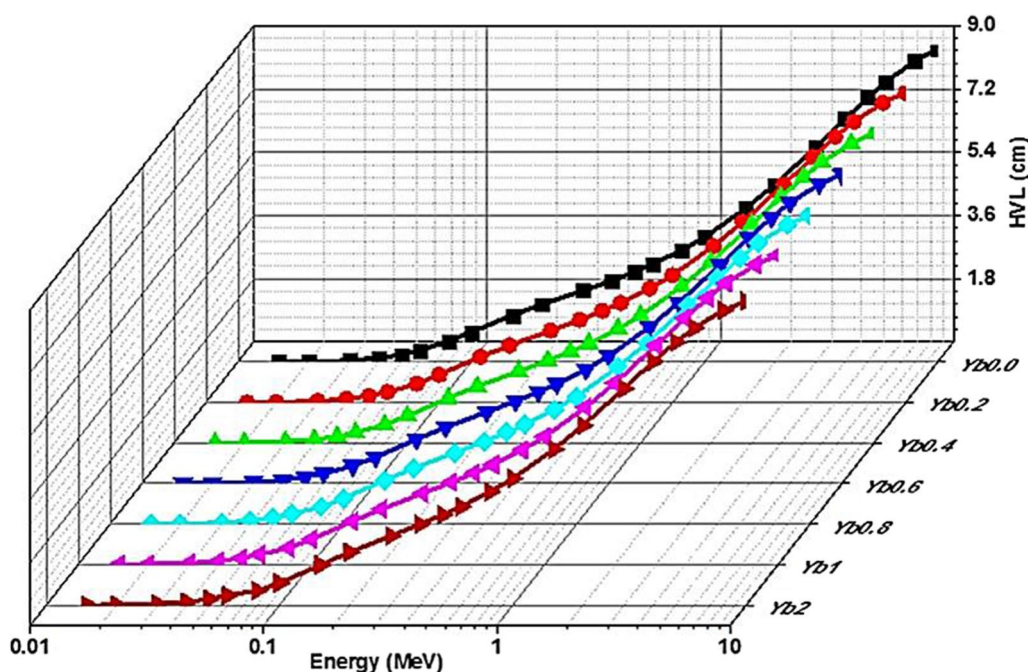


Fig. 6 HVL versus photon energy for all glass samples.

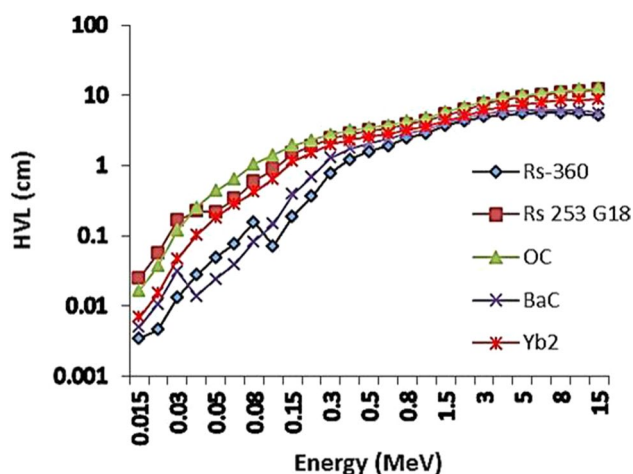


Fig. 7 Comparison of the HVL values of the Yb2 and commercial materials.

found to be $9.912 \times 10^{-21} \text{ cm}^2/\text{g}$, while for sample Yb2, it was found to be $13.06 \times 10^{-21} \text{ cm}^2/\text{g}$. The corresponding values of ECS were $(3.990 \text{ and } 4.900) \times 10^{-23} \text{ cm}^2/\text{g}$ for Yb0.0 and Yb2, respectively. It is clear that increasing the concentration of Yb_2O_3 in the prepared samples led to an increase in the atomic and electronic cross sections. In general, the values of the ACS were higher than the corresponding values of the ECS.

The effective atomic number (Z_{eff}) is a non-real parameter and varies with the energy of the incident photon. The effective atomic number is used to indicate the atomic number of composite materials, and it shows the shielding ability of the material. The Z_{eff} is also closely related to electron density N_{eff} and can be expressed as the number of electrons per unit mass. Figures 11 and 12 depict the behavior of Z_{eff} and N_{eff} for the present glasses versus the photon energy. Figure 10 we note that the Z_{eff} values for the present samples are found to be 24.83, 25.47, 25.60, 25.73, 25.87, 26.0 and 26.64 at 15 keV for Yb0.0–Yb2 glass samples, respectively. The highest Z_{eff} values were at low energies of 15 keV, after which a rapid decrease in the values of Z_{eff} until 0.1 MeV energy is obtained. One exception occurs at an energy of 63 keV, where a small peak appeared that was more pronounced for sample Yb2, and a shoulder peak at the same energy appeared for sample Yb1. This peak is due to the k-absorption edge of ytterbium. At energies from 0.1 to 2 MeV (Compton effect zone), the change in the Z_{eff} values was very small, then they slowly increased due to the dominant pair production in this region, which is proportional to Z^2 . Figure 12 shows the values of the electron density N_{eff} versus the photon energy. At an energy of 15 keV, the values of N_{eff} varied from 6.59×10^{23} to 6.50×10^{23} electrons/g for Yb0.0–Yb2, respectively. However, at this energy, N_{eff} is $(3.41\text{--}3.49) \times 10^{23}$ electrons/g for Yb0.0–Yb2, respectively. However, it is noticed that the behavior of N_{eff} is like that

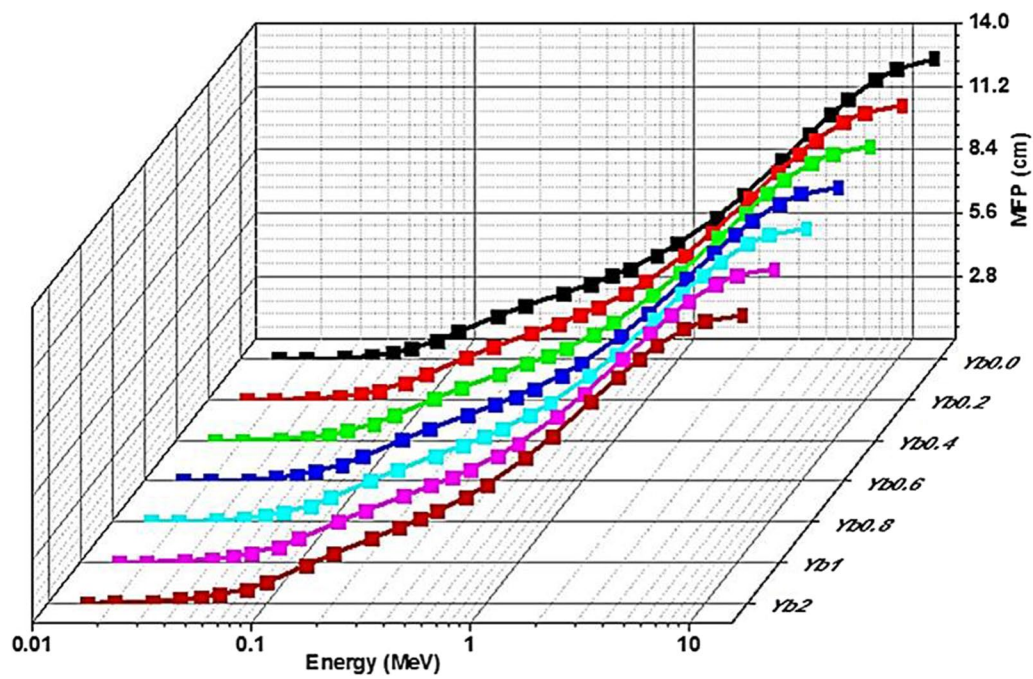


Fig. 8 MFP versus photon energy for all glass samples.

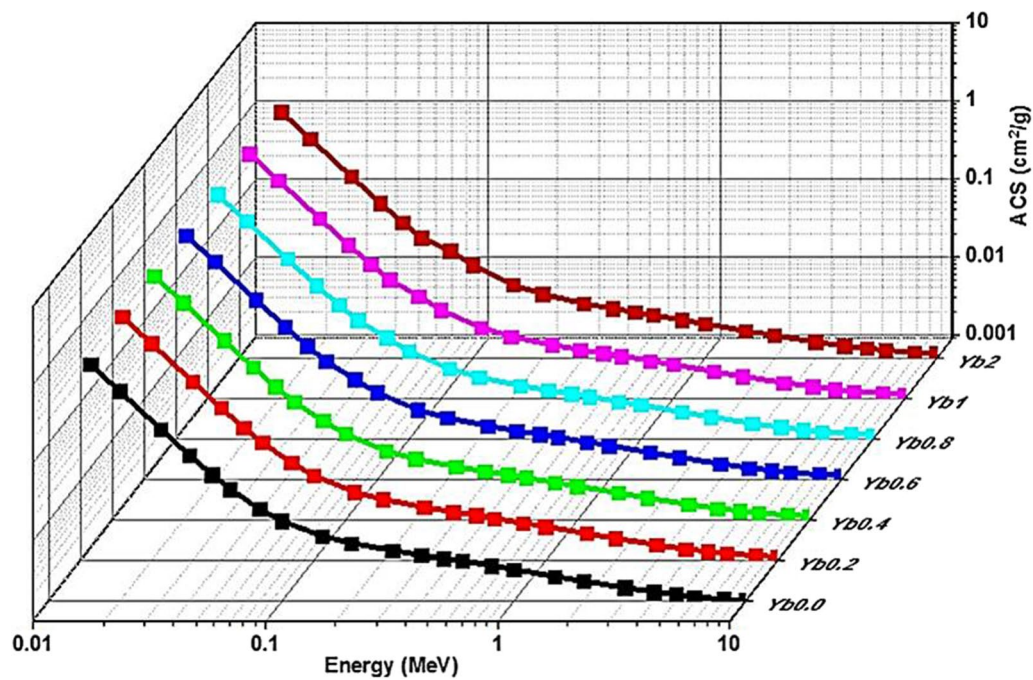


Fig. 9 Atomic cross section (ACS) versus photon energy for all glass samples.

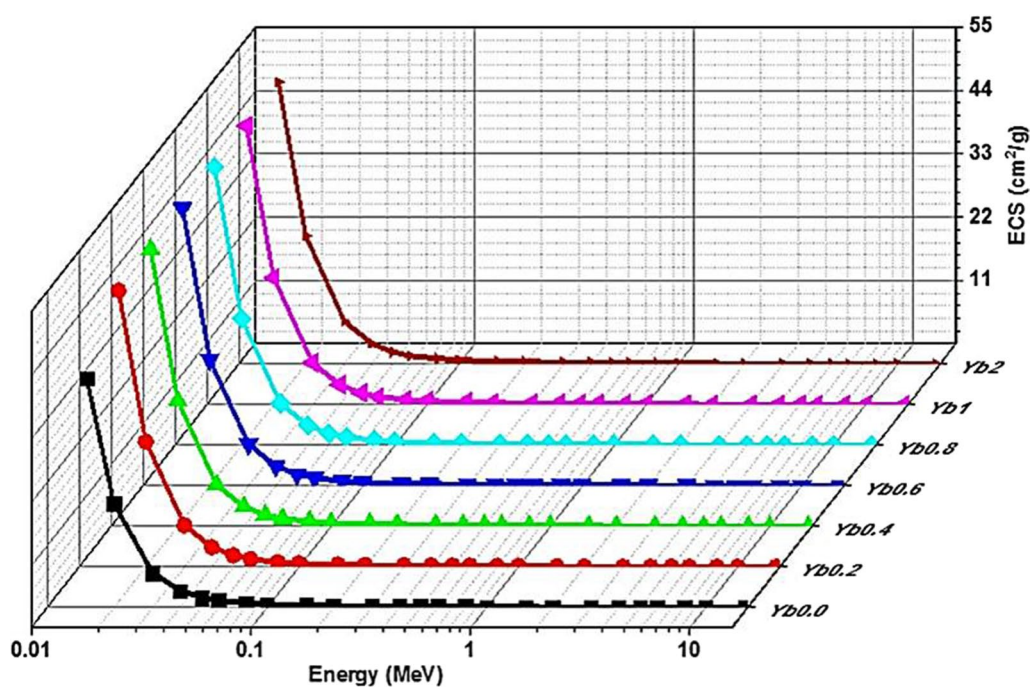


Fig. 10 Electron cross section (ECS) versus photon energy for all glass samples.

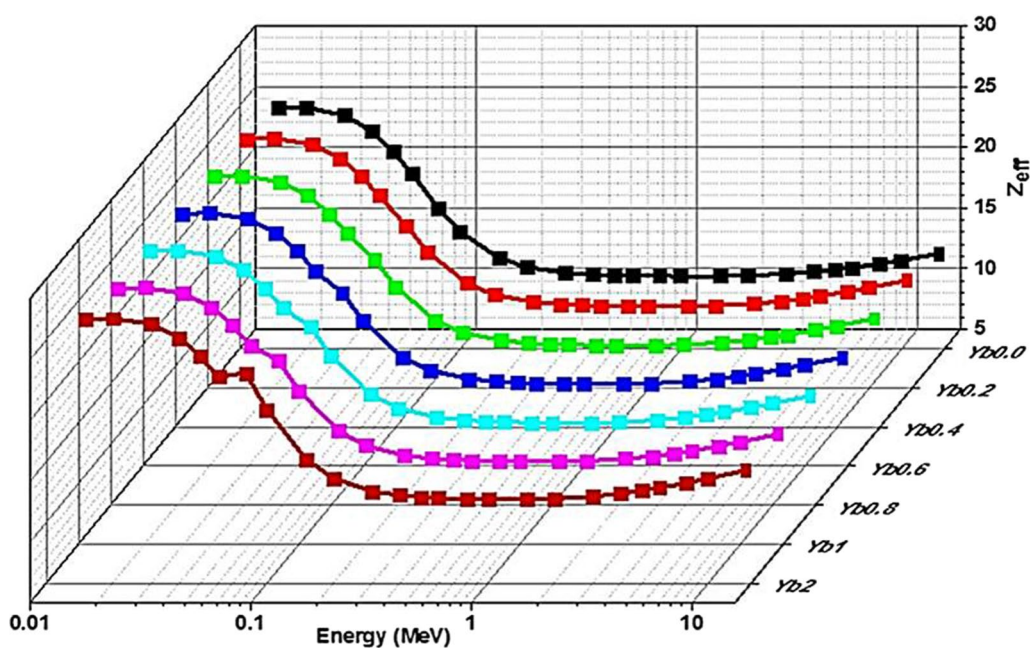


Fig. 11 Z_{eff} versus photon energy for all glass samples.

of Z_{eff} . Furthermore, the results of Z_{eff} of the present glass samples in comparison with Z_{eff} for concrete and commercial glasses and the results are present in Fig. 13. It can be noticed from the figure that the values of Z_{eff} for the Yb2 glass sample are higher than the corresponding values for

ordinary concrete and RS 253 G18 glass and lower than barite concrete and RS 360.

The buildup factor is mainly due to the accumulation of photons because of multiple scattering by the Compton effect. The energy absorption buildup factor (EABF) and the

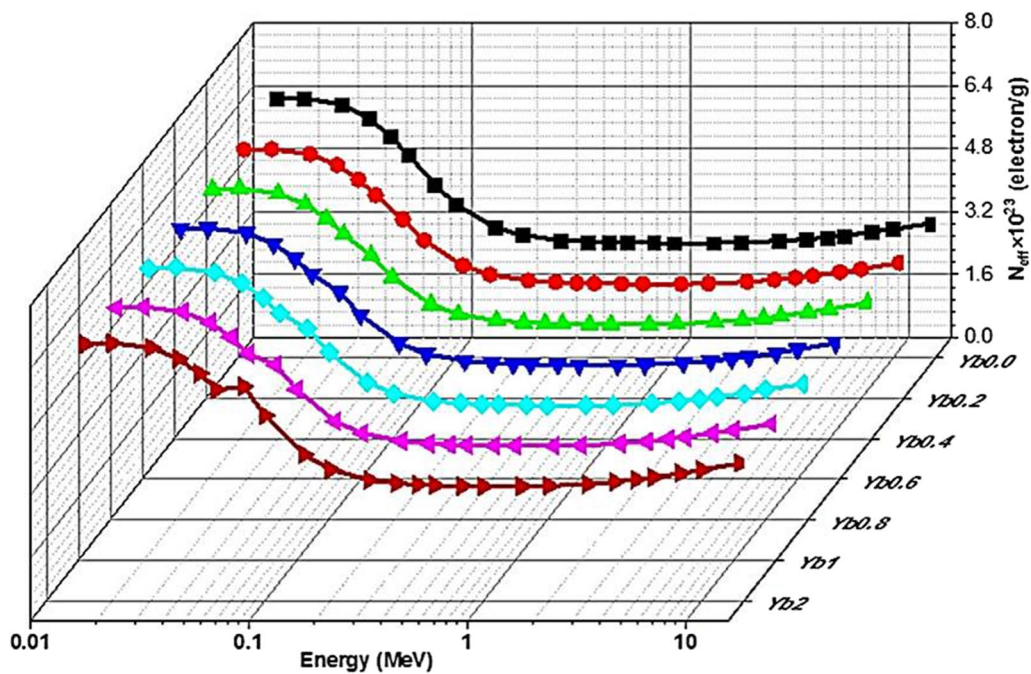


Fig. 12 N_{eff} versus photon energy for all present glass samples.

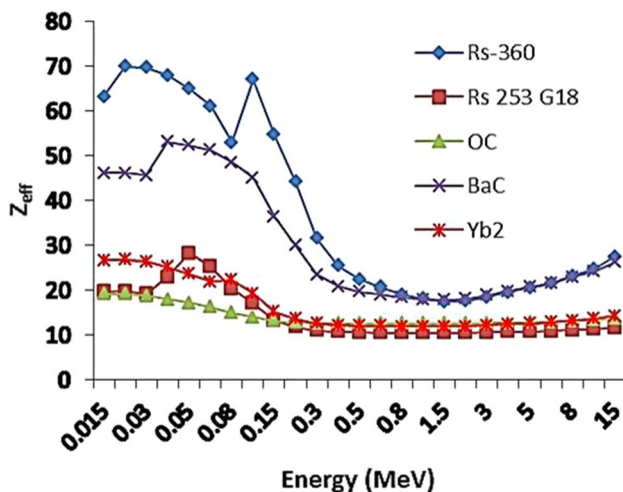


Fig. 13 Comparison of the Z_{eff} values of the Yb2 sample and commercial materials.

exposure buildup factor (EBF) are calculated using geometric progression (G-P) fitting parameters (a , b , c , X_k and d). The values of geometric progression (G-P) fitting parameters (a , b , c , X_k and d) for samples Yb0.0 and Yb2 are listed in

Tables III and IV. Using the Phy-X/PSD software program, EBF and EABF were obtained for prepared glass at 1, 10, 20, 30, and 40 mfp and the results are depicted in Figs. 14 and 15, respectively. According to the results in Figs. 14 and 15, the high values of EBF and EABF at gamma ray energy from 15 keV to 15 MeV were for sample Yb2. Moreover, the high values of EBF and EABF were at the moderate energy values, where the Compton effect is predominant in this region. The lowest values of EBF and EABF were in the low gamma energy region, where the photoelectric effect is dominant, because the electron absorbs all the energy of the incident photon. In the high-energy region, the values of EBF and EABF are moderate due to the high photon penetration into the glass, and in this region, pair production is predominant. Furthermore, from Figs. 14 and 15, it is noticeable that there is a slight increase in the EABF and EBF with the increase in the Yb_2O_3 contents. Also, it was observed that the obtained values of EABF are higher than those of EBF for all the studied glass samples.

The protective ability of material against uncharged particles (fast neutrons) is assessed by evaluating the macroscopic removal cross section (Σ_R) of the material. Large values of Σ_R mean that the number of interactions of neutrons

Table III EBF and EABF G-P fitting parameters (a, b, c, d and X_k) of the Yb0.0 glass sample.

Energy (MeV)	Z_{eq}	G-P fitting parameters for EBF					G-P fitting parameters for EABF				
		a	b	c	d	X_k	a	b	c	d	X_k
0.015	18.94	0.590	1.012	0.153	-0.596	12.504	0.405	1.010	0.241	-0.426	14.005
0.02	19.31	0.255	1.022	0.336	-0.126	11.915	0.328	1.021	0.286	-0.230	12.038
0.03	19.74	0.223	1.060	0.375	-0.117	11.757	0.239	1.059	0.360	-0.144	12.746
0.04	19.99	0.234	1.129	0.364	-0.136	14.068	0.222	1.127	0.380	-0.121	13.851
0.05	20.17	0.214	1.222	0.404	-0.118	14.136	0.226	1.236	0.388	-0.133	14.226
0.06	20.32	0.193	1.324	0.453	-0.111	14.245	0.192	1.369	0.445	-0.108	14.766
0.08	20.52	0.146	1.535	0.564	-0.082	14.363	0.141	1.702	0.560	-0.077	15.777
0.1	20.66	0.099	1.714	0.689	-0.061	14.276	0.109	2.128	0.660	-0.073	15.850
0.15	20.87	0.033	1.997	0.916	-0.031	13.760	0.067	3.111	0.824	-0.061	13.675
0.2	21.00	-0.003	2.112	1.079	-0.022	12.701	0.017	3.362	1.019	-0.039	12.933
0.3	21.14	-0.033	2.155	1.223	-0.013	10.597	-0.031	3.143	1.223	-0.016	10.717
0.4	21.21	-0.043	2.132	1.274	-0.010	9.975	-0.052	2.842	1.313	0.010	23.751
0.5	21.25	-0.046	2.091	1.289	-0.010	8.624	-0.059	2.618	1.344	0.006	15.474
0.6	21.28	-0.054	2.032	1.313	0.009	19.982	-0.059	2.479	1.335	0.010	18.578
0.8	21.31	-0.055	1.949	1.300	0.010	16.318	-0.059	2.263	1.321	0.009	14.301
1	21.31	-0.054	1.889	1.280	0.015	17.693	-0.057	2.129	1.292	0.015	16.831
1.5	18.42	-0.043	1.810	1.212	0.013	15.837	-0.046	1.941	1.222	0.014	15.010
2	16.37	-0.031	1.759	1.149	0.008	15.176	-0.030	1.838	1.148	0.007	15.749
3	15.72	-0.009	1.661	1.059	-0.005	11.090	-0.010	1.692	1.058	-0.003	11.720
4	15.55	0.008	1.588	0.999	-0.016	12.335	0.007	1.593	0.996	-0.013	12.887
5	15.48	0.015	1.518	0.973	-0.017	12.126	0.020	1.519	0.955	-0.027	14.593
6	15.43	0.019	1.469	0.958	-0.020	13.936	0.022	1.449	0.949	-0.029	15.404
8	15.37	0.030	1.389	0.928	-0.029	13.578	0.033	1.357	0.919	-0.029	12.256
10	15.35	0.037	1.328	0.911	-0.034	13.474	0.039	1.296	0.905	-0.036	13.605
15	15.32	0.051	1.233	0.885	-0.050	13.118	0.035	1.192	0.927	-0.035	14.409

with matter increases, and therefore, the number of fast neutrons decreases. The values of the (Σ_R) for the prepared glass samples were shown in Fig. 16. It can be noticed from the figure that the values of Σ_R were 0.103, 0.097, 0.097, 0.097, 0.096, 0.096 and 0.096 cm^{-1} for Yb0.0–Yb2, respectively. It should be noted that the values of Σ_R were decreased with the increase in Yb_2O_3 contents. This result is expected due to the replacement of Zn, Li and P by Yb in the network glass. Glass encoded Yb0.0 is more effective for capturing neutrons than other glass samples.

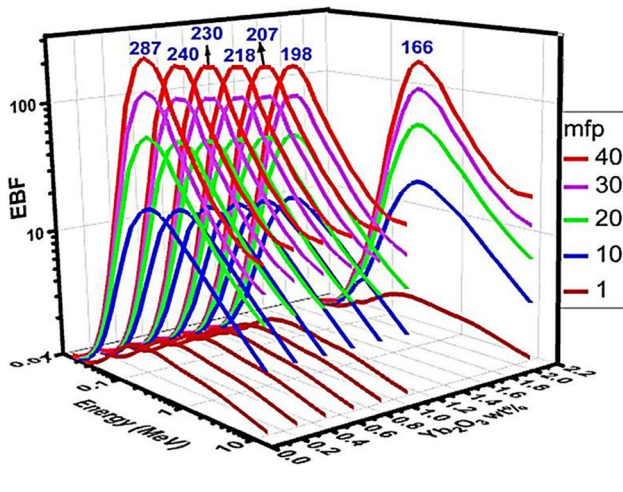
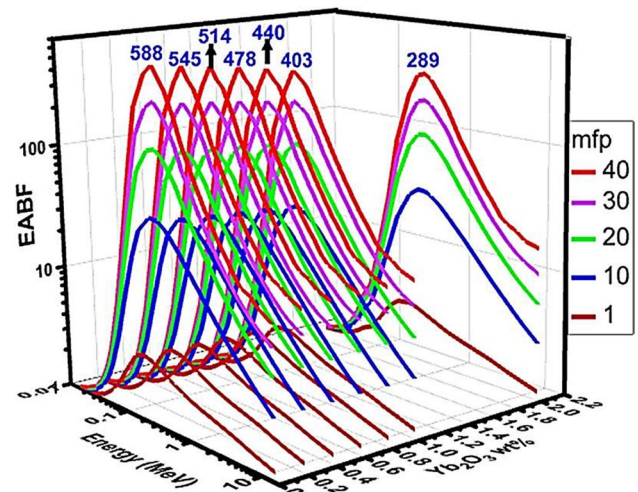
Conclusion

Seven glass samples were prepared according to 20 mol.% Li_2O –40 mol.% ZnO –40 mol.% P_2O_5 : x wt.% Yb_2O_3 ($x=0, 0.2, 0.4, 0.6, 0.8, 1, 2$). The samples were prepared by melt-quenching. The samples were examined by XRD and it was found that all the prepared samples were amorphous. The nuclear parameters of the prepared samples using XCOM and Phy-X/PSD programs were evaluated. The values

of MAC of the studied samples ranging from 26.034 to 31.872 cm^2/g for Yb0.0–Yb2, respectively. Gamma-ray parameters such as LAC, HVL, MFP, Z_{eff} , ACS, ECS, EABF and EBF were calculated. The HVL and MFP values ranged between 0.008–0.007 cm and 0.012–0.010 cm for Yb0.0–Yb2, respectively, while the ACS and ECS values obtained to be $9.912 \times 10^{-21} \text{ cm}^2/\text{g}$ to $13.06 \times 10^{-21} \text{ cm}^2/\text{g}$ and 3.990×10^{-23} – $4.900 \times 10^{-23} \text{ cm}^2/\text{g}$ for Yb0.0–Yb2, respectively. The values of Z_{eff} and N_{eff} were 24.83–26.64 and 6.59×10^{23} – 6.50×10^{23} electrons/g for Yb0.0–Yb2, respectively. It was concluded that the Yb2 glass sample was better than Yb0.0 as a gamma ray shield. The Σ_R was also obtained and found to be 0.103, 0.097, 0.097, 0.097, 0.096, 0.096 and 0.096 cm^{-1} for Yb0.0–Yb2, respectively, decreased with the increase of the Yb_2O_3 content. Moreover, the results of the current study were compared with the commercial materials used as gamma shields. Through the comparison, it was found that the Yb2 glass sample was better than ordinary concrete, as well as RS 253 G18 glass, while it was less efficient than barite concrete and RS 360 glass in terms of the protection against gamma rays.

Table IV EBF and EABF G-P fitting parameters (a, b, c, d and X_k) of the Yb2 glass sample.

Energy (MeV)	Z_{eq}	G-P fitting parameters for EBF					G-P fitting parameters for EABF				
		a	b	c	d	X_k	-0.010	1.007	0.658	0.129	9.412
0.015	20.40	-0.022	1.007	0.668	0.146	6.923	0.257	1.016	0.355	-0.193	12.212
0.02	20.73	0.240	1.015	0.398	-0.157	10.958	0.240	1.050	0.353	-0.156	14.100
0.03	21.15	0.218	1.051	0.372	-0.161	15.330	0.229	1.109	0.367	-0.124	13.863
0.04	21.41	0.238	1.111	0.357	-0.131	13.440	0.230	1.200	0.376	-0.133	14.273
0.05	21.60	0.219	1.189	0.394	-0.123	14.102	0.203	1.313	0.425	-0.116	14.748
0.06	21.76	0.197	1.275	0.440	-0.112	14.224	0.179	1.427	0.471	-0.101	15.271
0.08	24.56	0.172	1.332	0.493	-0.094	14.452	0.165	1.699	0.517	-0.099	15.159
0.1	24.90	0.136	1.450	0.582	-0.076	14.141	0.143	2.487	0.607	-0.113	14.866
0.15	25.47	0.075	1.692	0.759	-0.046	14.086	0.078	2.901	0.795	-0.071	13.114
0.2	25.84	0.033	1.847	0.916	-0.033	13.214	0.013	3.024	1.023	-0.037	12.503
0.3	26.28	-0.008	1.962	1.088	-0.018	11.924	-0.016	2.818	1.146	-0.023	11.779
0.4	26.56	-0.025	1.976	1.177	-0.014	10.863	-0.030	2.622	1.208	-0.015	10.637
0.5	26.74	-0.036	1.952	1.227	-0.008	8.744	-0.035	2.484	1.226	-0.012	10.403
0.6	26.86	-0.039	1.930	1.238	-0.009	8.298	-0.037	2.275	1.230	-0.013	8.474
0.8	26.99	-0.038	1.886	1.231	-0.011	7.623	-0.044	2.130	1.240	0.005	14.422
1	27.02	-0.043	1.832	1.234	0.005	15.247	-0.041	1.937	1.203	0.012	15.438
1.5	23.70	-0.041	1.766	1.202	0.012	15.854	-0.029	1.834	1.145	0.007	17.008
2	19.74	-0.030	1.739	1.147	0.007	17.295	-0.007	1.691	1.053	-0.006	12.797
3	18.20	-0.006	1.657	1.051	-0.009	11.549	0.012	1.592	0.986	-0.020	12.608
4	17.81	0.007	1.580	1.004	-0.015	10.901	0.017	1.503	0.969	-0.028	15.464
5	17.64	0.015	1.513	0.978	-0.020	13.191	0.026	1.439	0.946	-0.032	13.501
6	17.54	0.019	1.463	0.964	-0.023	13.648	0.033	1.344	0.926	-0.032	12.500
8	17.43	0.030	1.382	0.935	-0.033	13.401	0.040	1.283	0.910	-0.041	13.766
10	17.38	0.038	1.322	0.919	-0.039	13.510	0.035	1.180	0.940	-0.037	14.437
15	17.33	0.064	1.239	0.860	-0.063	13.668	-0.010	1.007	0.658	0.129	9.412


Fig. 14 Exposure buildup factor (EBF) versus photon energy for all glass samples.

Fig. 15 Energy absorption buildup factor (EABF) versus photon energy for all glass samples.

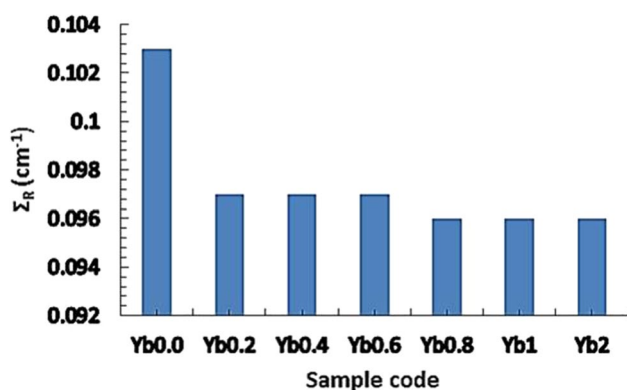


Fig. 16 Effective removal cross sections Σ_R for fast neutrons for all prepared glass samples.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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